

Stacked Bar Chart:

Under it the bars are horizontal. It is used for comparison analysis.

Example → I have taken four stacked bar chart. In it we are having category wise sales. We have dragged category in Y-axis & sales in X-axis.

* Click the Visual → Visualization pane → Format visual → Visual → our focus will be on (Y-axis, X-axis, Bars, Data labels).

* In general → effects → visual border and shadow options are there.

* In it everything is similar to first. Just we have to drag the ship date (year) in Legend.

Que → What is Legend?

Ans → It gives the detailed analysis or explanation.

Example → If I am having stacked bar chart of sales with region wise then if I want to see the sales year wise also then it will drag the year in the legend.

* Here our task was to green Bar for technology, Yellow bar for furniture and red for office supplies.

Select the visual → Visual → Bars → Categories (T, F, O) → Colour → Make the transparency zero%.

* In the bottom under the bar the border options is also there, we can border visualize.

* Sum of Sales by Category, year or segment under it, it is similar to the second visual. Just we have dragged segment in small multiples.

Que → What is small multiples?

Ans → It gives the **multiple visual** for a given visual with respect to particular criteria.

- Just we need to click on focus mode & then multiple visual with clear images will appear.
- The focus mode is near the **:** three dots when we click on any visual that time the three visual will come.

Que → How to create the duplicate?

Ans → Right click on the sheet name → Duplicate

→ Now we will create the three copies of this sheet in the first copy we will click the first visual → Visualization pane → Stacked column chart → we will do it with the remaining three charts.

→ In the next sheet we will use for clustered bar chart and clustered column chart.

Que → For what we use stacked and clustered chart?

Ans → We use stacked chart for **Contribution** and clustered chart for **Comparison**.

Suppose, In the second visual of sum of sales by category and year if I have to arrange the bars in descending orders with respect to

sum of sales
on **(:) three dots** → sort access → sort descending

Edit Interactions :

→ In PowerBI, the edit interactions tells that how one visual effects controls the effects on the remaining visual on the same report page.

Steps → Select the visual → format → click on edit interactions. After it on the remaining visual three options will be visible, filter, highlight, None.

Que → What is filter?

Ans → It shows the selected data and remove the remaining one.

Que → What is Highlight?

Ans → It keeps all the data visible but Highlight the selected ones.

100% Stacked Bar Chart :

→ In the stacked bar chart we see the contribution in numerical form but in 100% stacked bar chart we show the contribution in percentage form (in 100%).

100% Stacked Column Chart :

→ In the 100% stacked column chart shows how different facts make up a whole (100%) using percentage in each column.

Line Chart :

- The line chart shows the trend and we can compare the values also.
- I have dragged year in X-axis and sales and profit in Y-axis. Now I am able to see the trend but with it also able to compare sales and profit at a time.

Area Chart :

- Select the Line Chart → Format Visual → Shaded area on

Line chart = Area Chart

From the here, we conclude that line chart = area chart. Just we need to click on the shaded area. In the area chart, the area under the curves is shaded, otherwise it is completely line chart.

In short: Area chart shows trend and comparison but with shaded part and line chart do not.

Stacked Area Chart :

- Practically we observe that when I put the values in the legends in stacked area chart then it gives more classification as compared to area chart.
- It shows contribution in shaded part/area form.

Line and Stacked and Column Chart :

- Stacked column chart is used here to show. "Category wise sales" the contribution of stacked and column with respect to year. But if I put profit in line Y-axis then with the sales, it gives then with the sales it gives me information for the profit also.

- ② The measure-difference between line column chart & stacked column chart is the legends only.

Ribbon Chart:

- It is used to determine the ranks of the values.
 - It provide the rank. I have taken category in X-axis, sum of sales in Y-axis and years in legends.
- Now, if I click on Ribbon area then it takes me the change in percentage %. with in it given rank also.

Waterfall Chart:

- Under it we show the changes like rise and fall.
 - It shows the Rise and Fall.
- We taking year and Category and sales in Y-axis, and Quarter in breakdown. Now the green bar will show me the rise with increase percentage % and the red one shows the fall with decrease with percentage %.

Example → Suppose, if I am having the shop and having the sales of 200 in 1st year and 300 in 2nd year then this chart will show the change of +100.

It is having the option of breakdown also, if I put segment in Category and profit in Y-axis and put the years in breakdown then finally it will show the changes in value from year to year and one segment from other segment.

Funnel Chart :

- It shows me the output in funnel way means the Highest sales, at the top and then decrease in order to the overall. Picture is like Funnel.
- If I keep the cursor of any Bar then it shows the % with respect to the first value and with respect to the provides value.
- @ The highest value at the top, lowest value at the bottom. Just we need to sort the access in the descending order.
- @ The disadvantage under the funnel chart is that for knowing this all values we have to keep the cursor on regions that's why, it is optional chart.

Scatter Chart :

- It shows the comparison between two numerical columns.
Example → Profit, Sales

- Here we showing the relationship between profit and sales but we need to keep the region in the values. So, this chart will show the relation between sales and profit with respect to region.

Now, if I put the profit in the size then the size of the bubble according to the value of the profit.

Tree map:

Higher the value $\rightarrow$ Higher the size of square container in category (Country, details (region), value, & sales)

Now I am able to know in which country the "highest" or "lowest" sales are there.

Higher the value $\rightarrow$ higher the box/container
Lower the value $\rightarrow$ Lower the box/container
Smaller

Pie Chart:

$\rightarrow$ To show the composition of or proportion. It is generally preferred when the list is lesser than five.

Donut Chart:

$\rightarrow$ It is same as Pie Chart but it can consist the single KPI. Inside it center, firstly insert the donut chart $\rightarrow$ Insert the single card $\rightarrow$ Lets check & sales in the single card and keep it in the center of donut.

But the problem is coming when I click on anywhere on the donut chart, that time due to its background colour the single KPI gets hidden.

Steps

For remaining of this error select the donut chart
 $\rightarrow$ format your visual $\rightarrow$ general $\rightarrow$ effect
 $\rightarrow$ background off.

@ But we having advantages over the pie chart that under it, If we represented the sales with respect to segments then we can represented the total sales also, "In the center".

Stacked Column Chart :

- A stacked column chart is a graph that shows data using vertical bars, which shows contribution.
In simple terms, it shows how different facts contribute to a total.

Clustered Bar Chart :

- A clustered bar chart is used for comparison of different values across the same category.
It is in horizontal form.

Clustered Column Chart :

- Clustered column chart shows vertical bars and it is used for comparison of multiple values across the same category.

100% Stacked area chart :

- 100% stacked area chart shows how different facts show contribution from 100%. It helps to compare percentage contribution of each fact rather than actual values.

Line and Clustered Column Chart :

- A line and clustered column chart is a combination chart that shows columns and line together.
It helps to compare data and see the trends at the same time.